

// Question: Write a C program to draw a line using Bresenham's Algorithm.

```
#include <stdio.h>
#include <graphics.h>
#include <conio.h>
#include <math.h>
int main()
{
    int gd = DETECT, gm;
    int x1, y1, x2, y2, dx, dy, p, x, y;
    initgraph(&gd, &gm, "C:\\TURBOC3\\BGI");

    printf("Enter the starting coordinates (x1 y1): ");
    scanf("%d %d", &x1, &y1);
    printf("Enter the ending coordinates (x2 y2): ");
    scanf("%d %d", &x2, &y2);

    dx = abs(x2 - x1);
    dy = abs(y2 - y1);

    p = 2 * dy - dx;

    x = x1;
    y = y1;

    putpixel(x, y, RED);

    while (x < x2) {
        if (p >= 0) {
            y++;
            p = p + 2 * dy - 2 * dx;
        } else {
            p = p + 2 * dy;
        }
        x++;
        putpixel(x, y, RED);
    }

    getch();
    closegraph();
    return 0;
}
```